

Chapter 2: Review of Probability and Statistics Multiple Choice (with Solutions)

1. Problem

Continuous random variables

Which of the following random variables is a *continuous* random variable?

- (a) The time (in minutes) until the next customer enters a shop.
- (b) The payment method $\in \{\text{cash}, \text{card}\}$ of the next customer.
- (c) The number of defective items in a delivery of 100 items.
- (d) The result of rolling a die.
- (e) The number of customers who enter a shop within one hour.

Solution

- (a) True. Time can take any value in an interval, e.g., $[0, \infty)$.
- (b) False. Only two possible values: discrete (and not even numeric).
- (c) False. A count $\in \{0, 1, \dots, 100\}$ is discrete.
- (d) False. Only the six values $1, \dots, 6$: discrete.
- (e) False. A count $\in \{0, 1, 2, \dots\}$ is discrete.

2. Problem

Properties of a density function

Let f_X be the density function of a continuous random variable X . Which statement is correct?

- (a) $\int_{-\infty}^{\infty} f_X(x) dx$ can be any positive number.
- (b) $P(a \leq X \leq b) = f_X(b) - f_X(a)$ for all $a \leq b$.
- (c) $f_X(x)$ can take negative values, as long as $\int_{-\infty}^{\infty} f_X(x) dx = 1$.
- (d) $f_X(x) = P(X = x)$ for all x .
- (e) $f_X(x)$ can take values larger than 1.

Solution

- (a) False. The total area under a density is always exactly 1.
- (b) False. This formula holds for the cdf: $P(a < X \leq b) = F_X(b) - F_X(a)$. For the density, we need the area $\int_a^b f_X(x) dx$.
- (c) False. A density must be non-negative: $f_X(x) \geq 0$ for all x .
- (d) False. For a continuous random variable, $P(X = x) = 0$ for every single value x .
- (e) True. Density values are not probabilities; only areas under f_X are. For instance, $f_X(x) = 4$ for $0 \leq x \leq 0.25$ (and 0 otherwise) is a valid density.

3. Problem

Probabilities from a probability mass function

The number of complaints W per day has the probability mass function

w_k	0	1	2	3	4
$p_W(w_k)$	0.05	0.20	0.35	0.25	0.15

What is $P(W \geq 2)$?

- (a) 0.60
- (b) 0.75
- (c) 0.95
- (d) 0.35
- (e) 0.40

Solution

- (a) False. This is $P(W \leq 2)$.
- (b) True. $P(W \geq 2) = p_W(2) + p_W(3) + p_W(4) = 0.35 + 0.25 + 0.15 = 0.75$.
- (c) False. This is $P(W \geq 1)$.
- (d) False. This is only $P(W = 2)$.
- (e) False. This is $P(W > 2)$; the value $w = 2$ is missing.

4. Problem

Expected value of a discrete random variable

The number of items W that a customer buys has the probability mass function

w_k	1	2	3	4
$p_W(w_k)$	0.2	0.4	0.3	0.1

What is $E(W)$?

- (a) 2.30
- (b) 6.10
- (c) 0.58
- (d) 2.00
- (e) 2.50

Solution

- (a) True. $E(W) = \sum_k w_k p_W(w_k) = 1 \cdot 0.2 + 2 \cdot 0.4 + 3 \cdot 0.3 + 4 \cdot 0.1 = 2.30$.
- (b) False. This is $E(W^2) = \sum_k w_k^2 p_W(w_k)$, not $E(W)$.
- (c) False. The sum $\sum_k w_k p_W(w_k)$ must not be divided by the number of values; the probabilities are already the weights.
- (d) False. This is the most likely value (the mode), not the mean.
- (e) False. This is the simple average of the possible values 1, 2, 3, 4; it ignores the probabilities.

5. Problem

Probabilities from a cdf

The continuous random variable X has the cumulative distribution function

$$F_X(x) = \begin{cases} 0 & x < 0 \\ (x/5)^2 & 0 \leq x \leq 5 \\ 1 & x > 5. \end{cases}$$

What is $P(1 < X \leq 3)$?

- (a) 0.32
- (b) 0.40
- (c) 0.96
- (d) 0.16
- (e) 0.36

Solution

- (a) True. $P(1 < X \leq 3) = F_X(3) - F_X(1) = (3/5)^2 - (1/5)^2 = 0.32$.
- (b) False. This would be correct for a constant density on the interval, but here F_X is not linear.
- (c) False. This is $1 - F_X(1) = P(X > 1)$.
- (d) False. $F_X(3 - 1)$ is not the probability of the interval; use $F_X(3) - F_X(1)$.
- (e) False. This is $F_X(3) = P(X \leq 3)$; the probability $P(X \leq 1)$ must be subtracted.

6. Problem**Standard deviation of a linear transformation**

The monthly sales X (in units) of a product have standard deviation $SD(X) = 80$. The monthly profit (in EUR) is $\Pi = 15X - 1000$. What is $SD(\Pi)$?

- (a) 1,440,000
- (b) 1,200
- (c) 18,000
- (d) 200
- (e) 80

Solution

- (a) False. This is the variance $\text{Var}(\Pi) = 15^2 \cdot 80^2$, not the standard deviation.
- (b) True. $SD(aX + b) = |a| SD(X)$, thus $SD(\Pi) = 15 \cdot 80 = 1200$ EUR. The fixed costs do not affect the spread.
- (c) False. The factor must not be squared for the standard deviation (only for the variance).
- (d) False. The constant (fixed costs) shifts the distribution but does not change its spread.
- (e) False. Multiplying X by a factor also multiplies its standard deviation by that factor.

7. Problem**Variance of a linear transformation**

Let X be a random variable with $\text{Var}(X) = 5$. What is $\text{Var}(-2X + 10)$?

- (a) -10
- (b) 30
- (c) 10
- (d) 0
- (e) 20

Solution

- (a) False. The factor must be squared. (A variance can never be negative.)
- (b) False. A constant shift does not change the variance.
- (c) False. The factor must be squared.
- (d) False. The factor must be squared, and a constant shift does not change the variance.
- (e) True. $\text{Var}(aX + b) = a^2 \text{Var}(X) = (-2)^2 \cdot 5 = 20$.

8. Problem**Sample variance**

We observe $n = 4$ values: $X_{1,\text{obs}}, \dots, X_{4,\text{obs}} = 8, 7, 1, 4$. What is the sample variance $\hat{\sigma}_{X,\text{obs}}^2$?

- (a) 2.50
- (b) 30.00
- (c) 10.00
- (d) 7.50
- (e) 3.16

Solution

- (a) False. The sum of squared deviations must be divided by $n - 1$ only (once).
- (b) False. This is only the sum of squared deviations; it must be divided by $n - 1$.
- (c) True. $\bar{X}_{\text{obs}} = 5$ and $\hat{\sigma}_{X,\text{obs}}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_{i,\text{obs}} - \bar{X}_{\text{obs}})^2 = \frac{30}{3} = 10.00$.
- (d) False. The sample variance divides by $n - 1$, not by n .
- (e) False. This is the sample standard deviation $\hat{\sigma}_{X,\text{obs}}$.

9. Problem**Correlation**

Two random variables X and Y have $\text{Var}(X) = 16$, $\text{Var}(Y) = 25$ and $\text{Cov}(X, Y) = 15.00$. What is $\text{Cor}(X, Y)$?

- (a) 0.366
- (b) 0.600
- (c) 0.038
- (d) 0.750
- (e) 0.938

Solution

- (a) False. The denominator is $\sqrt{\text{Var}(X) \text{Var}(Y)}$, not the sum of the variances.
- (b) False. The covariance must be divided by both standard deviations.
- (c) False. The denominator is the *square root* of the product of the variances.
- (d) True. $\text{Cor}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{15.00}{\sqrt{16 \cdot 25}} = \frac{15.00}{20} = 0.750$.
- (e) False. The covariance must be divided by both standard deviations.

10. Problem

Covariance, correlation and units of measurement

For a sample of stores, the advertising expenditures X are measured in 1000 EUR and the sales Y in 1000 units. The sample covariance is $\widehat{\text{Cov}}_{\text{obs}}(X, Y) = 4$ and the sample correlation is $\widehat{\text{Cor}}_{\text{obs}}(X, Y) = 0.8$. Now, the advertising expenditures are measured in EUR instead (i.e., all X -values are multiplied by 1000). Which statement is correct?

- (a) The covariance becomes 4,000,000, and the correlation stays 0.8.
- (b) The covariance becomes 4000, and the correlation becomes 0.0008.
- (c) The covariance becomes 4000, and the correlation stays 0.8.
- (d) The covariance becomes 4000, and the correlation becomes 800.
- (e) The covariance stays 4, and the correlation stays 0.8.

Solution

- (a) False. Only X is rescaled, so the covariance is multiplied by 1000, not by 1000^2 .
- (b) False. The correlation does not change with the units.
- (c) True. $\text{Cov}(1000X, Y) = 1000 \text{Cov}(X, Y)$. The correlation does not depend on the units, since the factor 1000 cancels against the factor in $\text{SD}(1000X) = 1000 \text{SD}(X)$.
- (d) False. A correlation always lies in $[-1, 1]$; it does not change with the units.
- (e) False. The covariance depends on the units of measurement.

11. Problem

Conditional mean

The joint probability mass function $p_{Y,X}(y, x) = P(Y = y, X = x)$ of two discrete random variables $Y \in \{0, 1, 2\}$ and $X \in \{0, 1\}$ is given by

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.10	0.20	0.10
$X = 1$	0.30	0.20	0.10

What is the conditional mean $E(Y \mid X = 1)$?

- (a) 0.667
- (b) 0.600
- (c) 0.800
- (d) 1.000
- (e) 0.400

Solution

- (a) True. $p_X(1) = 0.30 + 0.20 + 0.10 = 0.60$, so the conditional pmf is $p_{Y|X}(y \mid 1) = 0.5, 1/3, 1/6$ for $y = 0, 1, 2$. Thus $E(Y \mid X = 1) = 0 \cdot 0.5 + 1 \cdot \frac{1}{3} + 2 \cdot \frac{1}{6} = \frac{2}{3} \approx 0.667$.
- (b) False. This is $p_X(1)$, the probability that $X = 1$.
- (c) False. This is the unconditional mean $E(Y) = 0 \cdot 0.4 + 1 \cdot 0.4 + 2 \cdot 0.2 = 0.8$.
- (d) False. This is $E(Y \mid X = 0)$.
- (e) False. $0 \cdot 0.30 + 1 \cdot 0.20 + 2 \cdot 0.10 = 0.4$ uses the joint probabilities; they must first be divided by $p_X(1) = 0.6$.

12. Problem
Independence

Let Y and X be two continuous random variables with joint density $f_{Y,X}$, marginal densities f_Y and f_X , and conditional density $f_{Y|X}$. Which statement is correct?

- (a) If $\text{Cor}(X, Y) = 0$, then Y and X are always independent.
- (b) Y and X are independent if $E(Y) = E(X)$.
- (c) If Y and X are independent, then $f_{Y,X}(y, x) = f_Y(y) + f_X(x)$.
- (d) If Y and X are independent, then $\text{Cor}(X, Y) = 1$.
- (e) If Y and X are independent, then $f_{Y|X}(y | x) = f_Y(y)$ for all y and x .

Solution

- (a) False. A correlation of zero only means that there is no *linear* association; a non-linear relationship may still exist.
- (b) False. Equal means say nothing about whether the variables are related.
- (c) False. Independence means that the joint density is the *product* $f_Y(y) \cdot f_X(x)$.
- (d) False. Independent random variables have correlation 0 (not 1).
- (e) True. Under independence, knowing $X = x$ does not change the distribution of Y .

13. Problem
Normal probabilities

Let $X \sim \mathcal{N}(100, 10^2)$ and let Φ denote the cdf of the standard normal distribution, with $\Phi(2.0) = 0.9772$. What is $P(X > 120)$?

- (a) 0.9545
- (b) 0.0455
- (c) 0.0228
- (d) 0.9772
- (e) 0.5000

Solution

- (a) False. This is $P(|X - 100| \leq 20)$.
- (b) False. This is $P(|X - 100| > 20)$, i.e., both tails.
- (c) True. $P(X > 120) = P\left(Z > \frac{120-100}{10}\right) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228$.
- (d) False. This is $\Phi(2.0) = P(X \leq 120)$.
- (e) False. Only half of the probability mass lies above the *mean*.

14. Problem
Chi-squared distribution

Let $Z_1, \dots, Z_3$ be independent standard normal random variables and $W = Z_1^2 + \dots + Z_3^2$. What is $\text{Var}(W)$?

- (a) 9.00
- (b) 2.00
- (c) 3.00

- (d) 6.00
- (e) 2.45

Solution

- (a) False. This is ν^2 ; the variance is 2ν .
- (b) False. The variance depends on the degrees of freedom: $\text{Var}(W) = 2\nu$.
- (c) False. This is the mean $E(W) = \nu$.
- (d) True. For $W \sim \chi_\nu^2$: $E(W) = \nu$ and $\text{Var}(W) = 2\nu = 6$.
- (e) False. This is the standard deviation $\sqrt{2\nu}$.

15. Problem**F-distribution**

Let $W_1 \sim \chi_3^2$ and $W_2 \sim \chi_{10}^2$ be independent, and let Z_1, Z_2 be independent standard normal random variables. Which random variable has an $F_{3,10}$ distribution?

- (a) $\frac{Z_1}{Z_2}$
- (b) $\frac{W_2/10}{W_1/3}$
- (c) $\frac{W_1}{W_2}$
- (d) $\frac{W_1 + W_2}{13}$
- (e) $\frac{W_1/3}{W_2/10}$

Solution

- (a) False. The F -distribution is built from chi-squared (i.e., squared normal) variables, not from normal ones.
- (b) False. This ratio has an $F_{10,3}$ distribution (the order of the degrees of freedom matters).
- (c) False. Both chi-squared variables must be divided by their degrees of freedom.
- (d) False. A sum of independent chi-squared variables is again chi-squared distributed (χ_{13}^2); dividing by 13 does not give an F -distribution.
- (e) True. The ratio of two independent chi-squared random variables, each divided by its degrees of freedom, is F -distributed; the numerator's degrees of freedom come first.

16. Problem**Regression function of a DGP**

The data are generated by $Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i$ with $\beta_1 = 40$, $\beta_2 = 2$, and $\varepsilon_i \sim \mathcal{N}(0, 8^2)$ independent of X_i . What is $E(Y_i | X_i = 10)$?

- (a) 42.0
- (b) 20.0
- (c) 60.0
- (d) 68.0
- (e) 402.0

Solution

- (a) False. The slope β_2 must be multiplied by x .
- (b) False. The intercept β_1 is missing.
- (c) True. $E(Y_i | X_i = 10) = \beta_1 + \beta_2 \cdot 10 + E(\varepsilon_i | X_i = 10) = 40 + 2 \cdot 10 + 0 = 60.0$.
- (d) False. The error term has mean zero; σ describes the spread around the regression line, not a shift.
- (e) False. The intercept and the slope are mixed up.

17. Problem**Law of Large Numbers**

Let $X_1, \dots, X_n$ be an i.i.d. random sample with mean $E(X_i) = \mu$. What does the Law of Large Numbers say?

- (a) For $n \geq 30$, the sample mean $\bar{X}_{\text{obs}}$ is exactly equal to μ .
- (b) For large n , the observations $X_{1,\text{obs}}, \dots, X_{n,\text{obs}}$ are approximately normally distributed.
- (c) The sample mean $\bar{X}_{\text{obs}}$ gets close to μ as the sample size n grows.
- (d) The variance of each X_i decreases as n grows.
- (e) For large n , every single observation $X_{i,\text{obs}}$ is close to μ .

Solution

- (a) False. For any finite n , the sample mean is an estimate and (almost always) differs from μ .
- (b) False. The distribution of the observations does not change with n . (The Central Limit Theorem is about the distribution of *sums/averages*, not of the single observations.)
- (c) True. $\bar{X}_{\text{obs}} \rightarrow E(X_i) = \mu$ as $n \rightarrow \infty$.
- (d) False. The distribution of each X_i does not depend on the sample size.
- (e) False. The single observations keep their spread $\text{Var}(X_i)$; only their *average* settles down.

18. Problem**Association and causation**

In a sample of primary school children, shoe size and reading skills are strongly positively correlated. Which statement is correct?

- (a) A strong correlation always implies a causal effect in at least one direction.
- (b) Since the correlation is strong, the relationship cannot be explained by a third variable.
- (c) The correlation is most likely driven by a confounder: older children have larger feet *and* read better.
- (d) Better reading skills cause larger feet.
- (e) Larger shoes cause better reading skills.

Solution

- (a) False. Association is not causation; a confounder can create a strong correlation without any causal effect between the two variables.

- (b) False. Confounders can create arbitrarily strong correlations.
- (c) True. Age influences both variables and creates the association (confounding). Among children of the same age, we would expect almost no correlation.
- (d) False. The correlation only shows a statistical association, not a causal effect.
- (e) False. The correlation only shows a statistical association, not a causal effect.